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5. The following information is about the Group 7 elements of the Periodic Table, called the halogens.

Group 7 – the halogens

H																	**He**
Li	**Be**											**B**	**C**	**N**	**O**	**F**	**Ne**
Na	**Mg**											**Al**	**Si**	**P**	**S**	**Cl**	**Ar**
K	**Ca**	**Sc**	**Ti**	**V**	**Cr**	**Mn**	**Fe**	**Co**	**Ni**	**Cu**	**Zn**	**Ga**	**Ge**	**As**	**Se**	**Br**	**Kr**
Rb	**Sr**	**Y**	**Zr**	**Nb**	**Mo**	**Tc**	**Ru**	**Rh**	**Pd**	**Ag**	**Cd**	**In**	**Sn**	**Sb**	**Te**	**I**	**Xe**
Cs	**Ba**		**Hf**	**Ta**	**W**	**Re**	**Os**	**Ir**	**Pt**	**Au**	**Hg**	**Ti**	**Pb**	**Bi**	**Po**	**At**	**Rn**
Fr	**Ra**																

The Group 7 elements all exist as diatomic molecules, meaning their atoms go around in pairs. Fluorine is found at the top of the group, astatine is at the bottom of the group.

Chlorine, atomic number 17, is a pale green gas at room temperature (20 °C). It has a melting point of −101 °C. Bromine, atomic number 35, is a red-brown liquid at room temperature. It has a melting point of −7 °C. Iodine, atomic number 53, is a grey solid at room temperature. It has a melting point of 114 °C.

Chlorine is used to make bleaches, to treat public drinking water and to sterilise public swimming pools. Bromine is used by gardeners in pesticides, to prevent insects from harming plants. Iodine solution is used as an antiseptic on wounds and on patients before being operated on.

(a) Which **one** of these statements best describes the trend in the melting points of the Group 7 elements as you go down the group? Tick (✓) the correct answer. [1]

The melting point increases ☐

The melting point decreases ☐

The melting point increases and then decreases ☐

There is no trend in the melting point ☐

(b) What is the correct formula for chlorine gas? Tick (✓) the correct answer. [1]

CL2		Cl_2		cl_2		2Cl	

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(c) Predict a value for the boiling point of chlorine and explain your answer. [2]

Boiling point of chlorine .... °C

Explanation ....

....

(d) Give the property that the Group 7 elements have that means they can be used as described in the text. [1]

....

(e) Decide whether the following statements about astatine are true or false. [2]

Astatine ...	True / False
... is a solid at room temperature	
... conducts electricity	
... has a melting point higher than 114 °C	
... is coloured	

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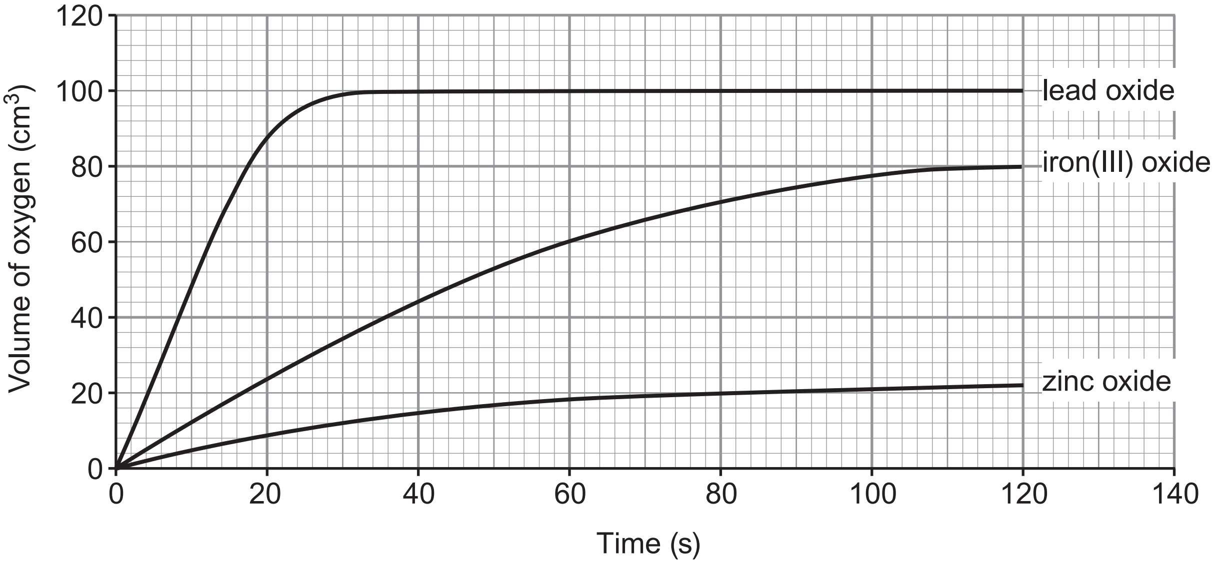
5. Hydrogen peroxide solution decomposes to give water and oxygen gas. Many metal oxides act as catalysts for the reaction.

(a) Use the symbol equation for the reaction to find the formula of hydrogen peroxide. [1]

$$2 \text{ ....} \longrightarrow 2H_2O \; + \; O_2$$

(b) Catalysts are used to speed up the reaction. The graphs show the volume of oxygen formed over 120 seconds when 1.0 g of different metal oxides were added to 50 cm^3 of hydrogen peroxide solution.



(i) Use the graph to find the time taken to form 44 cm^3 of oxygen using iron(III) oxide as the catalyst. [1]

.... s

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(ii) State which metal oxide is the **best** catalyst. Give a reason for your answer. [1]

....

....

....

(iii) After 120 seconds, the contents of each beaker were washed into a filter paper and funnel. The catalyst left over on each filter paper was dried and weighed.

Tick (✓) the box next to the statement which is correct. [1]

the same mass of all catalysts is left over ☐

more zinc oxide is left over than lead oxide ☐

about 80 % of the iron(III) oxide is left over ☐

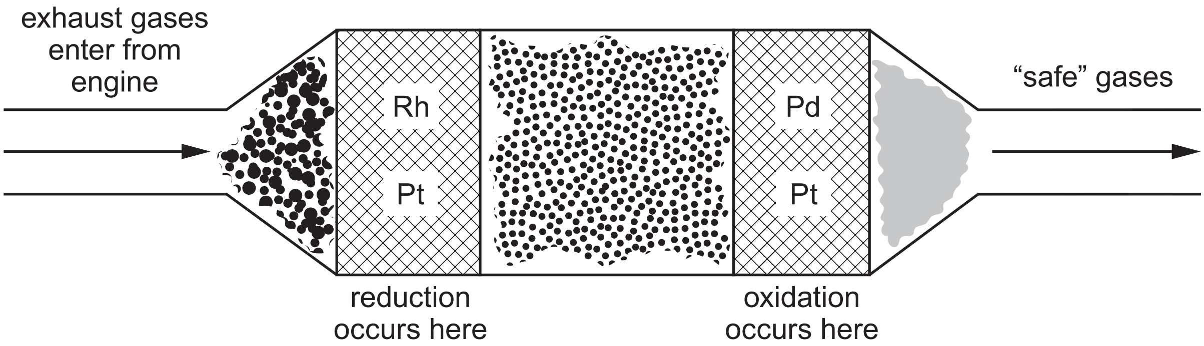
no lead oxide is left over ☐

(c) Exhaust gases from car engines contain harmful molecules, for example, carbon monoxide and nitrogen oxides. All cars are fitted with catalytic converters that split up these harmful molecules.

The catalysts are made from platinum (Pt) and palladium (Pd) or rhodium (Rh). A mesh structure is used that exposes the maximum surface area of catalyst to the exhaust gases, while also reducing the amount of catalyst required. Platinum, palladium and rhodium are extremely expensive.

As the exhaust gases from the engine pass over the catalysts, chemical reactions take place on their surfaces. The harmful molecules are broken up and converted into other gases that are "safe" to enter the air. These gases include carbon dioxide, nitrogen, oxygen and water.

Catalytic converter



There are two different types of catalyst, a **reduction catalyst** and an **oxidation catalyst**. The reduction catalyst removes oxygen from nitrogen oxides to make nitrogen and oxygen. The oxidation catalyst adds oxygen to carbon monoxide to make carbon dioxide.

One of the biggest drawbacks of the catalytic converter is that it only works effectively at high temperatures. The average running temperature of a car engine is between 90 and 110 °C. It takes nearly 30 minutes for these temperatures to be reached.

The table opposite shows the percentages of carbon monoxide and nitrogen oxides converted to safe gases by a catalytic converter at different temperatures.

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Temperature (°C)	Carbon monoxide converted (%)	Nitrogen oxides converted (%)
25	16	25
50	19	28
75	26	35
100	60	72
125	91	92
150	93	94
175	95	95
200	97	98

(i) Using your knowledge of particle theory suggest why a catalyst in mesh form works better than a lump of catalyst. [2]

....

....

....

....

(ii) Tick (✓) the box that best describes the adverse effect that gases leaving the exhaust would have on the environment. [1]

they have no effect on the environment ☐

they reduce oxygen levels ☐

they deplete the ozone layer ☐

they cause global warming ☐

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(iii) Tick (✓) the box that best describes the conversion of carbon monoxide and nitrogen oxides into "safe" gases at different temperatures. [1]

equal amounts of carbon monoxide and nitrogen oxides are converted at every temperature ☐

more carbon monoxide is converted than nitrogen oxides up to 100 °C ☐

more nitrogen oxides are converted than carbon monoxide up to 100 °C ☐

40 % more nitrogen oxides are converted than carbon monoxide up to 100 °C ☐

(iv) In your opinion how effective are catalytic converters? Explain your answer. [2]

....

....

....

....

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5. Planet **J** is similar in size to the Earth. However, the temperature on planet **J** is about 470 °C and the clouds in its atmosphere are made of sulfuric acid.

A group of students investigated the properties of some metals. Their aim was to see if they could find a metal suitable for designing a spacecraft to explore planet **J**. Their findings are shown below.

Zinc – fizzes quite vigorously with sulfuric acid, has a melting point of 420 °C and a density of 7.1 g/cm^3

Copper – has a melting point of 1083 °C, does not react with sulfuric acid and has a density of 8.9 g/cm^3

Sodium – has a density of 1.0 g/cm^3, reacts explosively with sulfuric acid and has a melting point of 98 °C

Magnesium – has a density of 1.7 g/cm^3, a melting point of 650 °C and it fizzes vigorously with sulfuric acid

Titanium – does not react with sulfuric acid, has a melting point of 1675 °C and a density of 4.5 g/cm^3

Lead – has a melting point of 328 °C, a density of 11.3 g/cm^3 and does not react with sulfuric acid

The spacecraft needs to withstand the conditions on the surface of planet **J**. The mass of the spacecraft also needs to be as low as possible in order for it to have enough energy to escape the Earth's gravity.

(a) Which **one** of these statements best describes why magnesium is an **unsuitable** metal for the spacecraft? Put a tick (✓) in the box next to the correct answer. [1]

its density is 1.7 g/cm^3 ☐

its melting point is 650 °C ☐

it fizzes vigorously with sulfuric acid ☐

it is malleable ☐

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(b) Small amounts of lead are sometimes used in electrical circuits.

Which **one** of these statements best describes why lead would **not** be suitable for use in the electrical circuits of the spacecraft? Put a tick (✓) in the box next to the correct answer. [1]

- it does not react with sulfuric acid ☐
- it is ductile ☐
- it would melt when it lands on planet **J** ☐
- its density is 11.3 g / cm^3 ☐

(c) The students decided that titanium is the most suitable metal from which to build the spacecraft.

Put a tick (✓) in the boxes next to the **two** statements that best describe the reasons for their choice. [2]

- it does not react with sulfuric acid ☐
- it is expensive ☐
- it is a good conductor of heat ☐
- it is non-magnetic ☐
- it has a melting point much higher than the temperature on planet **J** ☐
- it is shiny so will reflect the sun's rays ☐

(d) Sodium reacts explosively with sulfuric acid. Sodium sulfate and hydrogen are produced. Complete and balance the equation for this reaction. [2]

$$\Box\ Na + H_2SO_4 \longrightarrow \ldots\ldots\ldots\ldots\ldots\ldots + H_2$$

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5. Drinking water contains a number of ions which are important in the human body. Some of these ions cause hardness in water.

The table shows the concentration of ions in drinking water from four different locations.

Location	Concentration of ions (mg/dm^3 of water)					
	potassium K^+	ammonium NH_4^+	calcium Ca^{2+}	fluoride F^-	sulfate SO_4^{2-}	nitrate NO_3^-
A	0.1	0.4	0.0	0.0	0.4	0.2
B	0.0	0.3	0.4	4.4	0.2	0.0
C	0.2	0.6	2.7	0.4	0.0	0.1
D	3.4	2.1	1.0	2.1	2.5	2.3

(a) Which location is likely to have the hardest water?

Put a tick (✓) in the box next to the correct answer. [1]

A ☐

B ☐

C ☐

D ☐

(b) In which location do people have the least protection against tooth decay from their drinking water?

Put a tick (✓) in the box next to the correct answer. [1]

A ☐

B ☐

C ☐

D ☐

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(c) Ionic compounds dissolve in water to form positive and negative ions.

Which **two** compounds may have dissolved in the drinking water at location **B**?

Put a tick (✓) in the box next to the correct answer. [1]

potassium fluoride and calcium sulfate ☐

ammonium sulfate and potassium nitrate ☐

calcium fluoride and ammonium nitrate ☐

ammonium sulfate and calcium fluoride ☐

(d) (i) Calculate the relative formula mass (M_r) of calcium sulfate, $CaSO_4$. [1]

$A_r(Ca) = 40$ $A_r(S) = 32$ $A_r(O) = 16$

M_r = ...

(ii) Calculate the percentage by mass of fluorine in calcium fluoride, CaF_2. [2]

$M_r(CaF_2) = 78$ $A_r(F) = 19$

Percentage = ... %

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5. A teacher wanted to demonstrate the reactivity of lithium, sodium and potassium with water.

State what safety precautions the teacher should take and describe the observations you would expect for each metal.

You are **not** expected to include equations in your answer. [6 QER]

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